

A. MEX-Query

time limit per test: 2.5 seconds
memory limit per test: 512 megabytes

Well, you can't imagine what kind of a legend we could create for this problem. We wanted to create a long story on three pages with lots of unnecessary information. Another idea was to bind our task to some fictional situation, where two players play the game and they somehow already know Grundy values for their positions... But then we remembered **how much we dislike** such type of legends on contests, that have nothing common with the task, so all the spam in this statement ends in this paragraph.

MEX (*minimum excludant*) of a finite set of nonnegative integers S is the minimal non-negative integer $x \notin S$.

You are given a 1-indexed array A consisting of N nonnegative integers, and also Q queries (L_i, R_i) . For each query (L_i, R_i) , find a **MEX** for the subsegment of this array with indices from L_i to R_i inclusive.

Input

The first line contains N , the size of the array ($1 \leq N \leq 1\,000\,000$).

The second line contains N integers A_i ($0 \leq A_i \leq 1\,000\,000$) separated by spaces.

The third line contains Q , the number of queries you need to process ($1 \leq Q \leq 1\,000\,000$).

The following Q lines contain the description of the queries in the form $L_i\ R_i$ ($1 \leq L_i \leq R_i \leq N$).

Output

For each query according to their order in the input, print the **MEX**-value of the corresponding subsegment on a separate line.

Examples

input	Copy
10 2 0 1 3 0 0 2 4 7 1 7 1 2 5 6 5 10 1 3 1 5 6 6 7 8	
output	Copy
1 1 3 3 4 1 0	

2013-2014 Petrozavodsk Winter Training Camp, Moscow SU Trinity Contest

Finished

Practice

Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Submission	Time	Verdict
360790509	Jan/31/2026 03:04	Accepted

Contest materials

Statements (en)